

MYRTLE BEACH, SC, USA

WMO: 747910

Lat: 33.683N

Lon: 78.933W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13717

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.3	-13.9	1.1	-1.4	-11.2	1.4	1.7	10.9	9.7	9.6	7.7	2.7	350	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	32.8	25.4	32.0	25.5	31.0	25.3	27.3	30.6	26.6	30.0	25.9	29.3	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.7	29.3	25.7	21.0	29.0	24.9	20.0	28.3	85.8	30.5	82.6	29.7	80.4	29.6	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	8.4	7.9	-7.4	35.4	2.1	1.7	-8.9	36.6	-10.1	37.6	-11.2	38.5	-12.7	39.7
			-9.5	27.8	1.9	1.0	-10.9	28.6	-11.9	29.1	-13.0	29.7	-14.3	30.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	7.8	10.0	12.8	16.9	21.5	25.1	26.7	26.2	24.1	19.0	13.3	9.5	(6)
		DBStd	7.75	5.56	4.89	4.54	4.00	3.52	2.48	1.76	1.89	2.65	4.09	4.62	5.08	
		HDD10.0	286	111	56	26	1	0	0	0	0	1	19	73		
		HDD18.3	1307	328	236	176	73	12	1	0	2	43	160	276		
		CDD10.0	3125	42	56	113	208	357	452	518	502	422	279	119	57	
		CDD18.3	1105	1	2	6	30	111	203	259	243	174	63	10	2	
		CDH23.3	9432	1	6	22	126	747	1768	2649	2347	1370	361	31	4	
(13)		CDH26.7	2728	0	1	3	24	166	497	911	752	328	46	1	0	(13)
(14)	Wind	WSAvg	3.8	3.8	3.8	4.3	4.4	4.2	3.9	3.7	3.6	4.0	3.5	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	1295	90	89	96	70	93	121	147	170	159	103	72	84	(15)
(16)		PrecMax	1780	223	304	206	166	182	322	264	360	544	343	151	257	(16)
(17)		PrecMin	870	21	24	29	3	32	10	77	85	40	3	4	25	(17)
(18)		PrecStd	211	50	54	45	44	38	66	49	64	112	83	38	51	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	22.5	24.1	26.2	29.7	33.0	33.8	35.3	34.4	32.0	29.5	26.3	23.8		(19)
(20)		0.4% MCWB	17.2	17.7	17.6	19.1	23.5	24.8	26.7	26.2	24.6	24.2	20.2	18.2		(20)
(21)		2% DB	20.0	21.5	23.1	26.3	30.3	32.2	32.7	32.4	30.6	27.8	23.9	21.3		(21)
(22)		2% MCWB	16.3	16.7	16.9	19.3	23.3	24.8	26.1	26.2	24.7	22.8	19.4	17.4		(22)
(23)		5% DB	18.0	19.3	21.8	24.1	28.0	30.7	31.9	31.2	29.2	26.7	22.6	19.2		(23)
(24)		5% MCWB	15.1	15.9	16.6	18.7	22.3	24.6	26.0	25.8	24.4	22.2	18.7	16.7		(24)
(25)		10% DB	17.0	17.7	20.0	22.9	27.2	29.1	30.8	30.1	28.6	25.6	21.2	17.7		(25)
(26)		10% MCWB	14.3	14.6	15.7	18.0	21.8	24.2	25.7	25.3	24.0	21.7	17.9	15.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	18.6	19.6	20.2	21.7	25.3	27.1	28.4	28.3	26.7	25.5	22.2	19.6		(27)
(28)		0.4% MCDB	21.0	22.5	23.6	26.8	29.2	30.8	33.0	32.0	28.6	27.5	24.7	22.3		(28)
(29)		2% WB	17.0	18.1	18.7	20.8	24.4	26.1	27.4	27.3	25.8	24.5	20.9	18.3		(29)
(30)		2% MCDB	18.8	20.3	21.6	24.6	28.1	29.8	30.8	30.5	28.5	26.6	22.9	20.1		(30)
(31)		5% WB	16.0	16.8	17.6	20.0	23.6	25.5	26.8	26.5	25.3	23.5	19.6	17.2		(31)
(32)		5% MCDB	18.0	18.7	20.1	23.4	27.3	29.0	30.2	29.8	28.5	25.9	22.0	19.0		(32)
(33)		10% WB	14.6	15.3	16.6	19.1	22.8	24.8	26.1	25.8	24.7	22.4	18.3	15.7		(33)
(34)		10% MCDB	16.6	17.6	19.3	22.2	26.3	28.3	29.6	29.1	27.8	25.0	20.7	17.7		(34)
(35)	Mean Daily Temperature Range	MDBR	10.2	10.2	10.4	10.0	9.0	7.7	7.4	7.6	7.9	9.6	10.8	10.1		(35)
(36)		5% DB MCDBR	11.1	10.9	11.2	10.3	9.4	9.2	8.8	8.5	8.3	9.1	10.8	10.6		(36)
(37)		5% DB MCWBR	8.7	7.7	7.3	6.0	4.8	4.3	4.2	4.2	4.2	5.2	7.2	8.1		(37)
(38)		5% WB MCDBR	9.9	10.1	9.7	8.8	7.8	7.7	7.6	7.5	6.9	6.6	9.3	9.8		(38)
(39)		5% WB MCWBR	8.7	8.0	7.6	5.6	4.4	4.1	4.3	4.0	3.8	4.1	7.2	8.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.335	0.366	0.399	0.460	0.483	0.527	0.503	0.429	0.394	0.351	0.330		(40)
(41)		taud	2.522	2.498	2.431	2.326	2.208	2.189	2.084	2.147	2.346	2.414	2.490	2.542		(41)
(42)		Ebn at Noon	888	917	909	890	835	812	774	787	834	838	849	862		(42)
(43)		Edh at Noon	84	95	109	127	145	147	162	150	118	102	86	78		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.74	3.41	4.57	5.82	6.32	6.50	6.18	5.56	4.74	3.98	3.05	2.44		(44)
(45)		RadStd	0.17	0.34	0.41	0.39	0.43	0.42	0.38	0.31	0.36	0.43	0.23	0.19		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+118

Nomenclature: See separate page